Enerson (Livi) Poon.

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Objectives

- My mission is to keep patients out of the doctor's office by using artificial intelligence to discover and develop predictive, preventive, and precision medicine.
- Uncover problems and develop solutions for measurable phenomena (ie. anthropogenic processes — climate change) and their downstream outcomes (ie. ecosystem biology).
- Engage in ubiquitous environmental and scientific outreach, inspire current generations, and make measurable impacts toward a resilient, connected, and empathetic world!

Skills

- **Programming Languages**
  - Python.
  - Coursework in Java, C++, R, and Ruby.
  - Minor experience with JS, HTML, VueJS, NodeJS, and CSS.

- **Packages**
  - TensorFlow, PyTorch, Matplotlib, Pandas, Scikit-learn, and Seaborn.

- **Interests/Areas of Research**
  - Machine Learning, Artificial Intelligence, Neuroscience, Climate Change, Public Health, and Magnetic Resonance Imaging.

Research Experience

Paid Student Researcher - Karolinska Institutet - Institute of Environmental Medicine, Integrative Epidemiology Lab
May 2026 - Present

- Mapping disease progression pathways in GLP-1 receptor agonist users across Sweden's National Patient Registry and National Diabetes Registry using a novel 3D disease network framework (PheWAS, comorbidity networks, temporal trajectories).
- Leading a study to identify medicine biomarkers for ALS 1 year and 15 years before diagnosis using XGB, Swedish Registry data, SHAP, and LIME. In collaboration with the University of Helsinki and the University of North Carolina at Chapel Hill.
- Consulting and developing epidemiological machine learning studies with subject matter experts.

Joint Research Scholar - University of Vermont - Center for Complex Systems and Data Science, Computational Story Lab, and Neurobotics Lab
Aug 2023 - Present

- Analyzed U.S. Census and Department of Health & Human Services data using machine learning to identify statewide chronic disease mortality predictors and explore the relationship between socioeconomic factors and health trends. - Solo Project -
    - Research abstract and 2-page paper accepted to and presented at the 2025 Annual American Public Health Conference Meeting & Expo.
- Studied how body composition and psychosocial factors, such as self-compassion and social support, change across seasons in college students and identified key patterns that could improve mental health interventions and support.
    - Received a Summer Undergraduate Research Fellowship grant to continue research during the 2025 summer.
- Tested machine learning models and utilized Canonical Correlation Analysis to identify digital phenotypes and predict preterm and term Preeclampsia.
    - Research Abstract accepted and presented at the 2024 46th Annual International Conference of the IEEE Engineering in Medicine & Biology Society.
- Designed and tested pathways to expand undergraduate involvement in graduate-level research projects.
- Brainstormed new ways to improve data collection within MassMutual's data science team to advance rigorous, interpretable, and inclusive science for a global audience.
- Learned how to query and run jobs on UVM's computing cluster (supercomputer).

Machine Learning & Developmental Neuroscience Researcher - Stanford School of Medicine - Center for Interdisciplinary Brain Sciences Research, Dr. Hong Lab
Jun 2021 - Present

- Refining study methodology by implementing explainable AI methods (SHAP & LIME) and exploring their convergence.
- Developed machine learning models using scikit-learn and PyTorch to identify and compare brain regions that determine biological sex in structural magnetic resonance images of pre- and post-adolescents.
    - Presented research at Stanford School of Medicine's Maternal & Child Health Research Institute 2022 Symposium, Moderated Poster session.
- Manually corrected brain tissue segmentation in MRI images using FreeSurfer, an open-source software for processing and analyzing MRI images.
- Completed certification and operated 3T3 & 3T2 MRI machines at the Richard M. Lucas Center for Imaging.
- Performed functional connectivity analysis on resting-state fMRI images of Mosaic Turner Syndrome participants in the Puberty and Neurodevelopment in Adolescence (PANDA) study.

- Customized a Google Cloud virtual machine for remote data analysis and data acquisition.

- Developed a Bash CLI for FreeSurfer integration with Sherlock, Stanford's computing cluster (supercomputer).

Papers

- **Enerson Poon**, Hamed Honari, Javad S. Ansarifar, David S.Hong. (2026) Age stratification machine learning framework accurately captures neuroanatomical sex differences in adolescents. *Preprint coming soon on BioRxiv.*

- **Poon, E.**, Fudolig, M. I., Hidalgo, J. E., Loftness, B. C., Stanton, K., Tompkins, C. L., Bloomfield, L. S. P., Price, M., Dodds, P. S., Danforth, C. M., & Cheney, N. (2026). Identifying body composition measures that correlate with self-compassion and social support. *Manuscript accepted for publication in the Journal of American College Health. arXiv*. https://arxiv.org/abs/2602.18467

Conference Presentations

- **Enerson Poon**, Fen Yang, Yihan Hu, Elísabet Unnur Gisladottir, Can Hou, and Fang Fang: "Identifying disease trajectories within Swedish Type 2 Diabetes patients who discontinue GLP1 treatment", *IMM Research Symposium*, Stockholm, Nov 2026.

- Chris Danforth, Laura Bloomfield, Isabelle Smith, Peter Dodds, Mikaela Fudolig, Ryan McGinnis, Ellen McGinnis, Connie Tompkins, Johanna Hidalgo, Kathryn Stanton, Juniper Lovato, Taylor Ricketts, Mohsen Ghasemizade, **Enerson Poon** and Matthew Price: "Nature experiences improve mood and sleep in a large longitudinal study of well-being using the Oura ring", *IC2S2 12th Annual Conference on Computational Social Science*, Burlington, July 2026.

- **Enerson Poon**: "Preliminary Findings: Leveraging Machine Learning Models to Analyze Statewide Chronic Disease Measures that Predict Geographic Regions Across U.S. States," *UVM Student Research Conference*, Burlington, April 2026.

- **Enerson Poon**, Mikaela Fudolig: "Leveraging eXtreme Gradient Boosting and Multiclass Stratification to Analyze the Impact of Sociodemographic Factors on Statewide Chronic Disease Deaths Across U.S. States," *American Public Health Association*, Washington, D.C., November 2025. Abstract Link

- **Enerson Poon**, Mikaela Fudolig, Johanna Hidalgo, Bryn Loftness, Katheryn Stanton, Nick Cheney, Christopher M. Danforth, Connie Tompkins, Matthew Price: "Body Composition Correlates with Self-Reported Social Support and Self-Compassion Among College Students: Insights From the Lived Experiences Measured Using Rings Study (LEMURS)", *UVM Student Research Conference*, Burlington, April 2025.

- **Enerson Poon**, Bryn Loftness, Nick Cheney, Carole McBride, Ira Bernstein, "Selection Algorithms Identify Pre-Pregnancy Features Associated with Preeclampsia Onset and Birthweight Percentile", *Institute of Electrical and Electronics Engineers Engineering in Medicine and Biology Conference*, Orlando, Florida, July 2024.

- **Enerson Poon**, Javad Ansarifar, David S. Hong, "Deep Learning Identifies Sex-Specific Regions in Pre- and Post-adolescent Brains", *Stanford School of Medicine's Maternal & Child Health Research Institute 2022 Symposium*, Stanford, October 2022.

Professional Experience

Machine Learning & Data Science Intern - BETA Technologies - Data Team
May 2024 - Aug 2024

- Developed a machine learning pipeline using Scikit-Learn and Pandas to tag thermal runaway events in time-series pressure data. + GitLab CI for development, automated testing, and deployment.
- Improved software engineering skills: reusable code, docstrings, MyPy, and linting.
- Learned how to operate in a high-performance computing environment.

Outdoor Programs Intern - University of Vermont
Jul 2023 - Aug 2023

- Developed trip plan, gear lists, and interpersonal activities for a 23-person camping trip.
- Honed leadership, planning, outdoor recreation, outdoor education, and project management skills.
- Cleaned and reorganized 150+ lbs of medical supplies.

Software Engineering Intern - Loft Orbital
Feb 2021 - Aug 2021

- Developed and redesigned a 3D visualization component of Loft's satellite control software, displaying satellites, orbits, and ground stations in real time using Vite, Lit-Element, Vue.js, CSS, HTML, JS, Docker, and GitLab CI.
- Used Docker and GitLab CI for development, automated testing, and deployment.
- Improved leadership, project management, and team communication skills.

Awards

2026 - UVM - Presidential Leadership Fellowship

- *Worked with and learned from UVM President Marlene Tromp, Honors College Dean David Jenemann, and fellow industry leaders to understand the complexities of societal challenges, identify and investigate their impact across multiple scales, and examine and evaluate evidence-based solutions.*

2025 - UVM - Summer Undergraduate Research Fellowship

- *Received a financial fellowship to support further development of my psychosocial research on the Lived Experiences Measured Using Rings Study (LEMURS) over the summer of 2025.*

Personal Projects

PovellestoFilms (2020 - Present) (A Letter To Our Future Selves Short Film)

- Pursued a creative project to explore how to capture life authentically on camera.

PovellestoDraws (2023-2024)

- Started an art business, broke even, and made 7x initial investment by the end of month 1.
- Earned $600 in 3 hours by drawing portraits and selling original artwork

Certifications

May 2023 - SDI Open Water Scuba Diver (SDI)
May 2023 - Computer Nitrox Diver (SDI)
May 2023 - Reef Renewal Diver (PADI)

Education

2023 - Present - University of Vermont *(expected 2027)*

- Neurobotics & Artificial Intelligence (B.A.), Minor in Neuroscience
    - Intro to Machine Learning, Genetics, Molecular Bio, Psych Research Methods w/lab, Exploring Neuroscience w/lab, Learning, Cognition & Behavior

2018 - 2023 - The Nueva School *(HS Diploma - 2023)*

- Advanced Machine Learning, Computer Vision, Software Engineering, Algorithms, Computer Internals, Entrepreneurship, and Data Science.